Diagonalization and Powers of a Matrix

MIT 18.06SC, Session 22 — Worked Solutions

Problems 22.1 and 22.2

All arithmetic below is exact. Row operations shown in the solution are produced by SCAS.

Problem 22.1

Problem. Let $A = \begin{pmatrix} 4 & 0 \\ 1 & 2 \end{pmatrix}$. Diagonalize A, describe every possible diagonalizing matrix, and determine which matrices diagonalize A^{-1}.

Solution

Because A is triangular, its eigenvalues are visible on the diagonal. The characteristic polynomial confirms them:

$$p_A(\lambda) = \lambda^2 - 6 \cdot \lambda + 8$$

$$\begin{aligned} p_A(\lambda) &= 0 \\ &\Longrightarrow \lambda = 2, \quad \lambda = 4 \end{aligned}$$

For $\lambda = 4$, solving $(A - 4I)v = 0$ gives $v_1 = (2, 1)^{\mathsf{T}}$. For $\lambda = 2$, solving $(A - 2I)v = 0$ gives $v_2 = (0, 1)^{\mathsf{T}}$. SCAS obtains the same eigenbasis, listed by rows:

$$\mathcal{E}(A) = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \quad (\text{solve } (A - \lambda I)v = 0)$$

Put the eigenvectors in the columns of S, and put the corresponding eigenvalues on the diagonal of D:

$$S = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

To verify $A = SDS^{-1}$, first compute S^{-1} by Gauss–Jordan elimination, then multiply:

$$
\begin{aligned}
[S \mid I] &= \begin{pmatrix} 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \qquad \left(R_1 \leftarrow \frac{1}{2} R_1\right) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - (1)\, R_1) \\
S^{-1} &= \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
S\, D\, S^{-1} &= \begin{pmatrix} 8 & 0 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 0 \\ 1 & 2 \end{pmatrix}
\end{aligned}
$$

Hence $A = SDS^{-1}$. Since the eigenspaces are one-dimensional, every diagonalizing matrix is obtained by independently rescaling the two eigenvectors, with an optional swap of the columns. Equivalently,

$$
S' = \begin{pmatrix} 2a & 0 \\ a & b \end{pmatrix}, \qquad a, b \neq 0,
$$

or the same matrix with its two columns exchanged.

The inverse has the same eigenvectors and reciprocal eigenvalues. Thus

$$
\begin{aligned}
[A \mid I] &= \begin{pmatrix} 4 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 0 & \frac{1}{4} & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \qquad \left(R_1 \leftarrow \frac{1}{4} R_1\right) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{1}{4} & 0 \\ 0 & 2 & -\frac{1}{4} & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - (1)\, R_1) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{1}{4} & 0 \\ 0 & 1 & -\frac{1}{8} & \frac{1}{2} \end{pmatrix} \quad \left(R_2 \leftarrow \frac{1}{2} R_2\right) \\
A^{-1} &= \begin{pmatrix} \frac{1}{4} & 0 \\ -\frac{1}{8} & \frac{1}{2} \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
S\, D^{-1}\, S^{-1} &= \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{4} & 0 \\ -\frac{1}{8} & \frac{1}{2} \end{pmatrix}
\end{aligned}
$$

Therefore every matrix S' described above also diagonalizes A^{-1}, with diagonal matrix $\mathrm{diag}(1/4, 1/2)$ (or the swapped ordering).

Problem 22.2

Problem. Diagonalize the Markov matrix $B = \begin{pmatrix} 0.6 & 0.9 \\ 0.4 & 0.1 \end{pmatrix}$. Then find $\lim_{k\to\infty} \Lambda^k$ and $\lim_{k\to\infty} B^k$.

We use the exact fractions $3/5, 9/10, 2/5, 1/10$ in place of the printed decimals.

Solution

The trace and determinant provide a quick check on the characteristic polynomial:

$$\begin{aligned} \operatorname{tr}(B) &= \frac{3}{5} + \frac{1}{10} \\ &= \frac{7}{10} \end{aligned}$$

$$\begin{aligned} B &= \begin{pmatrix} \frac{3}{5} & \frac{9}{10} \\ \frac{2}{5} & \frac{1}{10} \end{pmatrix} \\ &\sim \begin{pmatrix} \frac{3}{5} & \frac{9}{10} \\ 0 & -\frac{1}{2} \end{pmatrix} \quad \left(R_2 \leftarrow R_2 - \left(\frac{2}{3}\right) R_1 \right) \\ \det(B) &= -\frac{3}{10} \qquad \text{(product of pivots, adjusted for row swaps)} \end{aligned}$$

$$p_B(\lambda) = \lambda^2 - \frac{7}{10} \cdot \lambda - \frac{3}{10}$$

$$\begin{aligned} p_B(\lambda) &= 0 \\ &\implies \lambda = -\frac{3}{10}, \quad \lambda = 1 \end{aligned}$$

The eigenvalues are 1 and $-3/10$. Corresponding eigenvectors are $(9, 4)^{\mathsf{T}}$ and $(1, -1)^{\mathsf{T}}$, so choose

$$\begin{aligned} Q &= \begin{pmatrix} 9 & 1 \\ 4 & -1 \end{pmatrix} \\ \Lambda &= \begin{pmatrix} 1 & 0 \\ 0 & -\frac{3}{10} \end{pmatrix} \end{aligned}$$

A row reduction gives Q^{-1}, and direct multiplication verifies the diagonalization $B = Q\Lambda Q^{-1}$:

$$
\begin{aligned}
[Q \mid I] &= \begin{pmatrix} 9 & 1 & 1 & 0 \\ 4 & -1 & 0 & 1 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & \frac{1}{9} & \frac{1}{9} & 0 \\ 4 & -1 & 0 & 1 \end{pmatrix} \qquad \left(R_1 \leftarrow \frac{1}{9} R_1\right) \\
&\sim \begin{pmatrix} 1 & \frac{1}{9} & \frac{1}{9} & 0 \\ 0 & -\frac{13}{9} & -\frac{4}{9} & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - (4)\, R_1) \\
&\sim \begin{pmatrix} 1 & \frac{1}{9} & \frac{1}{9} & 0 \\ 0 & 1 & \frac{4}{13} & -\frac{9}{13} \end{pmatrix} \quad \left(R_2 \leftarrow \frac{1}{-\frac{13}{9}} R_2\right) \\
&\sim \begin{pmatrix} 1 & 0 & \frac{1}{13} & \frac{1}{13} \\ 0 & 1 & \frac{4}{13} & -\frac{9}{13} \end{pmatrix} \quad \left(R_1 \leftarrow R_1 - \left(\frac{1}{9}\right) R_2\right) \\
Q^{-1} &= \begin{pmatrix} \frac{1}{13} & \frac{1}{13} \\ \frac{4}{13} & -\frac{9}{13} \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
Q\,\Lambda\,Q^{-1} &= \begin{pmatrix} 9 & -\frac{3}{10} \\ 4 & \frac{3}{10} \end{pmatrix} \begin{pmatrix} \frac{1}{13} & \frac{1}{13} \\ \frac{4}{13} & -\frac{9}{13} \end{pmatrix} \\
&= \begin{pmatrix} \frac{3}{5} & \frac{9}{10} \\ \frac{2}{5} & \frac{1}{10} \end{pmatrix}
\end{aligned}
$$

Since $B^k = Q\Lambda^k Q^{-1}$, the powers are controlled by

$$
\Lambda^k = \begin{pmatrix} 1 & 0 \\ 0 & (-3/10)^k \end{pmatrix}.
$$

The second diagonal entry tends to zero. Therefore

$$
\lim_{k\to\infty} \Lambda^k = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.
$$

Substitution gives the limiting projection

$$
\lim_{k\to\infty} B^k = Q \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} Q^{-1} = P = \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix}.
$$

As an exact check, the reduced row-echelon form has one pivot, so P has rank one:

$$
\begin{aligned}
P &= \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 1 \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix} \quad \left(R_1 \leftarrow \frac{1}{\frac{9}{13}} R_1 \right) \\
&\sim \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad \left(R_2 \leftarrow R_2 - \left(\frac{4}{13} \right) R_1 \right) \\
\operatorname{rref}(P) &= \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}
\end{aligned}
$$

Finally, P is stationary and idempotent:

$$
\begin{aligned}
B\,P &= \begin{pmatrix} \frac{3}{5} & \frac{9}{10} \\ \frac{2}{5} & \frac{1}{10} \end{pmatrix} \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix} \\
&= \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix} \\
P^2 &= \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix} \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix} \\
&= \begin{pmatrix} \frac{9}{13} & \frac{9}{13} \\ \frac{4}{13} & \frac{4}{13} \end{pmatrix}
\end{aligned}
$$

Thus every initial probability vector converges to the steady-state vector $(9/13, 4/13)^{\mathsf{T}}$.